



The forecasting and policy analysis systems (FPAS) macro-model for Rwanda

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Abstract

In view of the challenges hampering the effectiveness of monetary policy embedded in the monetary targeting regime, the BNR is in the process of switching to the inflation targeting framework where the role of modeling and forecasting is paramount. Towards this end, the BNR has taken significant steps to put in place a Forecasting and Policy Analysis Systems (FPAS). This paper has developed and applied an FPAS macro-model for Rwanda focusing on explaining the key episodes in Rwanda's macroeconomic developments. The FPAS macro-model is structured in such a way that it captures the key transmission channels. Understanding these key transmission channels is crucial for it sheds light on how the Rwandan economy works, which itself is a useful input in the production of forecasts for key macro-economic variables. The model results adequately explain the historical developments in GDP growth, inflation dynamics and exchange rate movements between 2008 and 2014-15. However, the model needs to be extended to better capture the intrinsic realities of the Rwandan economy by for example including the fiscal sector, incorporating the policy actions, especially exchange rate smoothing, and the role of money. The good performance of the current model and the expected future improvements will strengthen the operationalization of the model to produce forecasts for key macroeconomic variables aimed at guiding monetary policy discussions. Currently, some of the model results, such as the projections on the output gap and inflation, are used to inform the Monetary Policy Committee.

Key Words: Forecasting & Policy Analysis systems, macro-model, Rwanda

JEL Classification: E52, E58, F47, O23

1. Introduction

During various policy discussions, the Central Banks across the East African Community (EAC) have acknowledged the numerous challenges limiting the effectiveness of monetary policy in the region. In the International Research conference held in Kigali on 19th-20th July 2012, Central Bank Governors in the region acknowledged the weak monetary policy transmission mechanism ingrained in the monetary targeting regime. The monetary targeting framework, which is generally based on the quantity theory of money, is based on simple and unrealistic assumptions. First, it assumes a stable and predictable relationship between money supply and inflation, holding money velocity constant. Indeed, for a stable relationship between money supply and inflation to exist, the money multiplier must be stable. Recent empirical evidence shows that there is growing breakdown of the link between money supply and inflation in the EAC region, largely due to the instability of the money multiplier (Adam and Kessy, 2011; Kigabo and Irankunda, 2012). In this respect, the Central Bank Governors committed themselves to modernize monetary policy regimes and this has repeatedly featured in the East African community's Monetary Affairs Committee (MAC) meetings.

At present, the EAC central banks have embarked on the modernization of their monetary policy frameworks with the objective of transiting to a more price-based forward looking monetary policy framework by 2018. Some of the changes introduced so far include, allowing greater exchange rate flexibility, enhancing the role of policy rates in the signaling of the monetary policy stance, announcing inflation targets and introducing the forward-looking elements in policy formulation and communication strategies (Luisa Charry et al., 2014).

The EAC Central Bank Governors in the region agreed to develop and operationalize the Forecasting and Policy Analysis Systems (FPAS), which has various forward-looking elements entrenched in the Central Bank structure and policy processes. These forward-looking elements include: putting in place a team fully dedicated to forecasting with clearly defined responsibilities, developing a database infrastructure with a system that collects and organizes the relevant information for this exercise, introducing a set of near-term forecasting and nowcasting tools, building a quarterly

projection model, conducting a cycle of meetings on a regular basis to update the forecast and interact with senior management, and adopting a reporting process that presents the analysis in a coherent, clear and straightforward manner so as to facilitate policy discussions.

In the FPAS context, the National Bank of Rwanda has already taken significant steps that will provide a firm foundation for the envisaged shift to a forward-looking monetary policy. The reforms undertaken include the on-going modernization of electronic systems (including the putting in place of a data ware house), revision of the monetary policy framework whereby a reserve money band is targeted, using of the policy rate to signal the monetary policy stance, improving communication of monetary policy decisions to the public (including publication of inflation reports and MPC press releases), setting up of the division in charge of modeling and forecasting, the continued use of near-term forecasting tools (such as ARMA and VARs) and most recently the ongoing efforts to build and fully operationalize the small quarterly projections model (QPM). Other reforms include the introduction of measures to absorb excess liquidity and improve the inter-bank market as well as the BNR commitment to maintain low and stable inflation, not exceeding 5%.

The QPM under the FPAS is simply a small Keynesian macro-economic model, with four core equations describing developments in inflation (Phillips curve), aggregate demand (IS curve), exchange rates (UIP) and the behavior of monetary authorities (Taylor rule) as explained in Luisa Charry et al. (2014).

This paper explains the structure and application of the QPM to the Rwandan economy. The main advantages of the QPM are: the ability to model the macro-economic linkages between sectors, the latitude to incorporate expert judgment (calibrations), and the provision of the room for having forecasts over a longer time horizon as well as serving as a consistent framework for streamlining monetary policy discussions, among others.

The successfulness of the model to explain key historical developments of the Rwandan economy is a stepping stone towards production of reliable forecasts for key macro-economic variables, such as for inflation and output gap, that the MPC needs



for policy discussions. Therefore, the main focus of this paper is to use the QPM to explain the key episodes in Rwanda's macroeconomic developments. Periods such as 2008, 2011, 2014-15 were adequately explained by the QPM to especially understand the dynamics behind inflation developments. Some of these explanations were provided in previous studies such as Luisa Charry et al. (2014) and Andrew Berg et al. (2013).

The rest of the paper is organized as follows: section one gives the introduction with particular emphasis on the background, structure of and rationale for the FPAS model; section two, which is the gist of this paper, dwells much on the application of the FPAS model to the Rwandan economy and gives a model consistent interpretation of the Rwandan economy over the recent past by especially explaining the inflation dynamics in 2008, 2011 and 2014-15. The third section dwells on the main conclusions of the study. The interpretations are based on the filtration results for the last eleven years, since 2004. The aforementioned QPM is cast in state-space format and filtered using the Kalman filter.

2. The FPAS macro-model for Rwanda

2.1 The model structure

The FPAS macro model upon which this paper is based is simply a rational expectations New-Keynesian model, similar to models used in central banks around the world, with both nominal and real rigidities. It is specified in gaps and the main mechanism driving inflation over the business cycle is fluctuations in real variables, such as output, around their long-term trends. The model is basically suitable for a fully-fledged inflation targeting country with a flexible exchange rate. Therefore, the main policy instrument is the interest rate whereas the long-term neutrality of money is implied.

The model consists of four basic behavioral equations: the IS curve (aggregate demand), which relates monetary policy and the real economic activity; a set of Phillips curves (aggregate supply) that link economic activity and inflation; a monetary policy

rule that describes the response of the central bank to deviations of inflation from target and the cyclical behavior of the economy.

2.1.1 The aggregate demand (IS curve) equation

As explained by Ferrara (2008), Giannone (2005) and Billi (2012), the most important measure of economic performance is real GDP. Under the FPAS framework, the output gap is used as a proxy for aggregate demand and is measured as percentage deviation of actual output from its potential level¹¹. The rationale is that a positive output gap implies overheating of the economy and upward pressures on inflation. Conversely, a negative output gap implies economic slowdown and downward pressures on inflation. The influences of the output gap on inflation are well captured in the Phillips curve. The aggregate demand equation is specified as follows:

$$\hat{y}_t = a_1^{\hat{y}} * \hat{y}_{t-1} - a_2^{\hat{y}} * rmc_t + a_3^{\hat{y}} * \hat{y}_{t-1}^{US} + \varepsilon_t^{\hat{y}} \dots \dots \dots (1)$$

Whereby:

$\hat{y}_t$ is the output gap, rmc_t is the real monetary condition index; $\hat{y}_{t-1}^{US}$ is the lagged global aggregate demand, approximated here using the lagged US output gap, and; $\varepsilon_t^{\hat{y}}$ is the demand shock.

The output gap is computed as the difference between actual ($l_{-}y_t$) and potential output ($l_{-}\bar{y}_t$), measured in logs:

$$\hat{y}_t = l_{-}y_t - l_{-}\bar{y}_t \dots \dots \dots (2)$$

Where by $l_{-}y_t = 100 * \log(RGDP_t)$ and the gap and trend are derived using the Kalman filter; and $RGDP_t$ stands for real GDP for quarter “t”.

¹¹ For details about the estimation of the output gap in Rwanda, see Kigabo and Irankunda (BNR, economic review, 2014, N°5 pp. 19-40).

foods imported to be used as final goods, can be sourced from the rest of the world and are thus not excluded when computing core inflation.

Conversely, both imported food and oil inflation are included in the non-core equation since Rwanda's non-core basket contains energy and fresh foods. Common to both equations however is the fact that imported oil and food inflation represent the direct effect, usually seen as a temporary but immediate spike in inflation emanating from an increase in either international food or oil prices. Since international oil and food prices are quoted in USD, they have to be converted to the FRW before being entered into core and non-core equations respectively. The real marginal costs, which capture the indirect effects to domestic inflation, are specified as:

$$rmc_t^{core} = b_1^{rmc_core} * l_y_gap_t + b_2^{rmc_core} * (l_qfood_gap_t + l_z_gap_t + w_1^{rmc} * \hat{r}p) + b_3^{rmc} * (l_qoil_gap_t + l_z_gap_t + w_1^{rmc_core} * \hat{r}p) + (1 - b_1^{rmc_core} - b_2^{rmc_core} - b_3^{rmc_core}) * (l_z_gap_t + (w_1^{rmc} - 1) * \hat{r}p) \dots (11a)$$

$$rmc_t^{xcore} = b_1^{rmc_xcore} * l_y_gap_t + b_2^{rmc_xcore} * (l_qfood_gap_{t-1} + l_z_gap_{t-1} + w_1^{rmc} * \hat{r}p_{t-1}) + b_3^{rmc_xcore} * (l_qoil_gap_t + l_z_gap_t + w_1^{rmc_xcore} * \hat{r}p) + (1 - b_1^{rmc_xcore} - b_2^{rmc_xcore} - b_3^{rmc_xcore}) * (l_z_gap_t + (w_1^{rmc} * \hat{r}p_{t-1}) \dots (11b)$$

Whereby, $l_qfood_gap_t$ is the real food price gap, $l_qoil_gap_t$ is the real oil price gap, $\hat{r}p$ is the relative prices gap whereas other variables are named as before. Therefore $\hat{r}p$ is the relative prices gap and w_1^{rmc} partitions the effect of relative prices between core and non-core inflation equations, respectively. The inflation expectations are modeled as follows:

$E\pi_t^{core}$ & $E\pi_t^{xcore}$, standing for core and non-core inflation forward-looking expectations, respectively. Since the core and non-core inflation equations contain $E\pi_{t-1}$ and $E\pi_{t+1}$, they capture both forward-looking and backward-looking inflation expectations and must also satisfy the linear homogeneity assumption.

Having two Phillips curves, for core and non-core inflation, brings forth the need to tie together these two measures of inflation, especially due to the fact they do not grow at the same rate. To establish this model-consistent relationship, the idea of relative prices between core and non-core inflation is used:

$$l_rp_t = l_P_t^{core} - l_P_t^{xcore} \dots\dots\dots(12)$$

Where l_rp_t is the real relative price between core and non-core CPI; $l_P_t^{core}$ is 100 times log of core CPI; and, $l_P_t^{xcore}$ is 100 times the log of non-core CPI. The real relative price is then decomposed into trend and gap and the growth rate in the trend is, after making the necessary changes, modeled as a stationary process, in a similar way as in equations (28) and (34). With core and non-core inflation processes explicitly modeled, headline inflation is then derived as an identity using the respective weights of core and non-core inflation as follows:

$$\pi_t^{headline} = wt^{core} * \pi_t^{core} + (1 - wt^{core}) * \pi_t^{xcore} \dots\dots\dots(13)$$

2.1.3 The uncovered interest rate parity (UIP) equation

The uncovered interest rate parity equation explains how the nominal exchange rate (s_t) is determined. The nominal exchange rate is expressed in RWF/USD terms. According to equation 15 below, dynamics of the nominal exchange rate depend on expectations one quarter ahead (Es_{t+1}), nominal interest rate differentials, defined as the difference between the foreign nominal interest rate (i_t^*) and the domestic nominal interest rate (i_t), and the country's risk premium ($prem_t$). The interest rates are measured in annual terms. The premium is the compensation required by foreigners for holding domestic securities.

$$Es_{t+1} = d_1 * Es_{t+1} + (1 - d_1) [s_{t-1} + 2/4 * (dl_z_t + \pi_t^{4,tar} - \pi_{ss}^*)] \dots\dots\dots(14)$$

$$s_t = Es_{t+1} + (i_t^* - i_t + prem_t)/4 + \varepsilon_t^s \dots\dots\dots(15)$$

The expected one quarter ahead nominal exchange rate, taken as a weighted average between backward and forward-looking expectations, is approximated by a combination of the model consistent forecast for the nominal exchange rate and the backward looking expectation of the nominal exchange rate. The backward looking

term is then one quarter lagged nominal exchange rate level (S_{t-1}) plus the equilibrium nominal depreciation over the two periods ahead. The equilibrium nominal depreciation is given by equilibrium real exchange rate depreciation dl_z and the difference between the domestic inflation target ($\pi_t^{A,tar}$) and foreign steady state inflation (π_{ss}^*) as shown in equation 14.

In addition, the real (long run) UIP assumes parity relationship between expected equilibrium real exchange rate appreciation ($e_1_dl_z_t$), differential of equilibrium real interest rates $\bar{r}_t - \bar{r}_t^*$ and the country risk premium ($prem_t$). This equality implicitly pins down the country risk premium. This specification of the UIP does not however take into account the possibility of central bank intervention in the forex market mainly aimed at smoothing exchange rate volatility. The latter constitutes one of the areas to be considered for further improvements of the FPAS model in future.

$$\bar{r}_t = \bar{r}_t^* + e_1_dl_z_t + prem_t \dots \dots \dots (16)$$

$$\bar{r}_t - \bar{r}_t^* = e_1_dl_z_t + prem_t \dots \dots \dots (17)$$

$$e_1_dl_z_t = dl_z_{t+1} \dots \dots \dots (18)$$

2.1.4 The monetary policy reaction function

The monetary policy reaction function does not necessarily imply a central bank rule but rather a model representation of how the central bank conducts monetary policy. The monetary policy rule explained here is typically for inflation targeting countries using the interest rate as an operational target. Though the National Bank of Rwanda uses reserve money as the operational target, the key repo rate is up to now used as a signal of the monetary policy stance until full transition to the inflation targeting regime is attained. The monetary policy rule is specified in a way to reflect the main objectives of the central bank. It is here assumed that the central bank sets the nominal interest rate in response to the deviations of the one year a head inflation forecast from the inflation objective and the output gap.

The long run objective of monetary policy is to bring inflation to the objective. Inflation deviation from the target is defined as one- quarter-ahead model consistent forecast of y-o-y inflation minus the inflation target. The first short run objective is to smooth the output gap and keep the economy close to its potential level. Another objective is to smooth nominal exchange rate volatility with respect to the trend implied by real exchange trend and difference in inflation targets. Therefore monetary policy is consistent and keeps the exchange rate in accordance with its primary target, which is the medium-term inflation objective. The resulting Taylor rule, taking into account all these objectives is specified as follows:

$$i_t = f_1 * i_{t-1} + (1 - f_1) * [i_neutral + f_2 * \pi_t^{yoy,dev} + f_3 * \hat{y}_t + f_4 * dl_s_t^{qoq,dev}] + \varepsilon_t^i \dots \dots \dots (19)$$

The monetary policy reaction function stated in equation (19) implies that the inter-bank rate, used here as a proxy for the policy rate (i_t) depends on the persistence term (i_{t-1}), the neutral interest rate ($i_neutral$), deviation of YOY inflation from the target ($\pi_t^{yoy,dev}$), the output gap ($\hat{y}_t$) and QOQ annualized deviation of the nominal exchange rate depreciation from the trend ($dl_s_t^{qoq,dev}$) as well as the monetary policy shocks (ε_t^i). The neutral nominal interest rate is defined in equation (20) as a sum of the real interest rate trend ($\bar{r}_t$) and the two periods ahead annual inflation target (equations 20 and 21):

$$i_neutral_t = \bar{r}_t + E\pi_t^{yoy,tar} \dots \dots \dots (20)$$

$$E\pi_t^{yoy,tar} = \pi_{t+2}^{yoy,tar} \dots \dots \dots (21)$$

The YOY inflation deviation from the target is given by:

$$\pi_t^{yoy,dev} = E\pi_t^{yoy} - \pi_t^{yoy,tar} \dots \dots \dots (22)$$

The deviation of the nominal exchange rate depreciation from the target is defined as:

$$dl_s_t^{qoq,dev} = dl_s_t - (dl_z_t + \pi^{yoy,tar} - ss_pi^*) \dots \dots \dots (23)$$

Whereby dl_{s_t} is the QoQ annualized nominal exchange rate appreciation/depreciation, $dl_{\bar{z}_t}$ is equilibrium real exchange rate depreciation, $\pi^{voy, tar}$ is the annual inflation target and ss_{π^*} is the steady state level of foreign inflation. The term in brackets therefore gives the equilibrium nominal exchange rate depreciation. The neutral interest rate is defined as in equation (20) while the expected QOQ annualized inflation target is defined in equation (21). Note that the nominal interest rate is the closure rule for the model and ensures convergence to the steady state position.

2.1.5 Oil prices

Oil price quoted here is the international oil price measured in USD, converted into a real price in equation (25) using the US CPI index as the deflator. That is, the real price of oil is modeled as a relative price of oil to the foreign CPI. Then, equation (26) splits the real price of oil into a trend and gap. When real oil price is at trend (or equilibrium) level, then international oil prices have no pressure to other domestic prices. Equation (30) shows how the QoQ annualized % real oil price trend growth rate is computed while equation (29) explains the computation of the annualized Q-o-Q oil inflation. Equations (27) and (28) show the processes for the QoQ annualized % real oil price trend growth rate and the real oil price gap, respectively.

$$\Delta l_{oil_im_t} = \Delta l_{oil_t} + \Delta l_{s_t} - \Delta l_{z_tnd_t} \dots \dots \dots (24)$$

$$l_{qoil_t} = l_{oil_t} - l_{P_t^*} \dots \dots \dots (25)$$

$$l_{qoil_t} = l_{qoil_tnd_t} + l_{qoil_gap_t} \dots \dots \dots (26)$$

$$\Delta l_{qoil_tnd_t} = c_1^{\Delta l_{qoil_tnd}} \Delta l_{qoil_tnd_{t-1}} + \varepsilon_t^{\Delta l_{qoil_tnd}} \dots \dots \dots (27)$$

$$l_{qoil_gap_t} = c_1^{l_{qoil_gap}} l_{qoil_gap_{t-1}} + \varepsilon_t^{\Delta l_{qoil_gap}} \dots \dots \dots (28)$$

$$\Delta l_{oil_t} = 4 * (l_{P_t^{oil}} - l_{P_{t-1}^{oil}}) \dots \dots \dots (29)$$

$$\Delta l_{qoil_tnd_t} = 4 * (l_{qoil_tnd_t} - l_{qoil_tnd_{t-1}}) \dots \dots \dots (30)$$

Equation (27) implies that the equilibrium real oil price is modeled as a stationary process such that the innovation to this shock process moves the level of the real price

of oil only slowly to a new permanent level. The shock to this stationary process corresponds to the technology innovation process in the world economy. The gap of the real oil price represents all other movements in the real price of oil which imply inflation pressures. This gap is also modeled as a stationary AR (1) process in equation (28).

2.1.6 Food prices

The international food price is modeled in similar fashion as the international oil prices with the only exception that the steady state of the trend of the equilibrium of real price of food is equal to zero. This is premised on the fact that food takes a significant part of Rwanda's CPI basket, with a share of about 28%.

$$\Delta l_food_im_t = \Delta l_food_t + \Delta l_s_t - \Delta l_z_tnd_t \dots \dots \dots (31)$$

$$l_qfood_t = l_food_t - l_P_t^* \dots \dots \dots (32)$$

$$l_qfood_t = l_qfood_tnd_t + l_qfood_gap_t \dots \dots \dots (33)$$

$$\Delta l_qfood_tnd_t = c_1^{\Delta l_qfood_tnd} * \Delta l_qfood_tnd_{t-1} + \varepsilon_t^{\Delta l_qfood_tnd} \dots \dots \dots (34)$$

$$l_qfood_gap_t = c_1^{l_qfood_gap} * l_qfood_gap_{t-1} + \varepsilon_t^{\Delta l_qfood_gap} \dots \dots \dots (35)$$

$$\Delta l_food_t = 4 * (l_P_t^{food} - l_P_{t-1}^{food}) \dots \dots \dots (36)$$

$$\Delta l_qfood_tnd_t = 4 * (l_qfood_tnd_t - l_qfood_tnd_{t-1}) \dots \dots \dots (37)$$

2.1.7 Long run trends

The FPAS macro-model aims at explaining the medium-term business cycle dynamics of the economy around its long-term trends. There are various ways of estimating long-term trends but the most commonly used one is the Kalman Filter technique, which estimates all the unobservable variables in the model structure and imposes restrictions on volatility. Below are the key trends estimated by the model:

$$\bar{r}_t = \phi_1 * \bar{r}_{t-1} + (1 - \phi_1) * \bar{r}_{ss} + \varepsilon_t^{\bar{r}} \dots \dots \dots (38)$$

$$\bar{z}_t = \phi_2 * \bar{z}_{t-1} + (1 - \phi_2) * \Delta \bar{z}_{ss} + \varepsilon_t^{\Delta \bar{z}} \dots \dots \dots (39)$$

$$\Delta \bar{y}_t = \phi_3 * \Delta \bar{y}_{t-1} + (1 - \phi_3) * \Delta \bar{y}_{ss} + \varepsilon_t^{\Delta \bar{y}} \dots \dots \dots (40)$$

2.1.8 The foreign sector

The dynamics of the model are complete with the foreign block. Though exogenous to the domestic economy, the developments in the foreign sector tend to affect the domestic economy for example through the presence of the foreign output gap in the domestic IS equation, the effect of the foreign nominal interest rate on the domestic economy's UIP equation and the role of foreign inflation on the domestic economy's real exchange rate developments.

The US economy is used here as the foreign sector. Other options include using both the US and Eurozone since most of the trade is conducted using the dollar and euro as mediums of exchange. This option is more complicated since it requires the use of relative weights. The other option is consideration of the major EAC trading partners. This is equally more complicated since it not only requires weighting but also suffers from data problems since actual data and forecasts for developing countries are always not readily available on a timely basis. The foreign sector foreign block contains equations for the US: output gap (equation 39), real interest rate trend (equation 40), nominal interest rate (equation 41) and headline inflation (equation 42). These equations are modeled as stationary processes rather being dependent on macro-economic fundamentals since the model takes the foreign block as given.

$$\hat{y}_t^{US} = a_1^{\hat{y}^{US}} * \hat{y}_{t-1}^{US} + \varepsilon_t^{\hat{y}^{US}} \dots\dots\dots(41)$$

$$\bar{r}_t^{US} = a_2^{\bar{r}^{US}} * \bar{r}_{t-1}^{US} + (1 - a_2^{\bar{r}^{US}}) * \bar{r}_{ss}^{US} + \varepsilon_t^{\bar{r}^{US}} \dots\dots\dots(42)$$

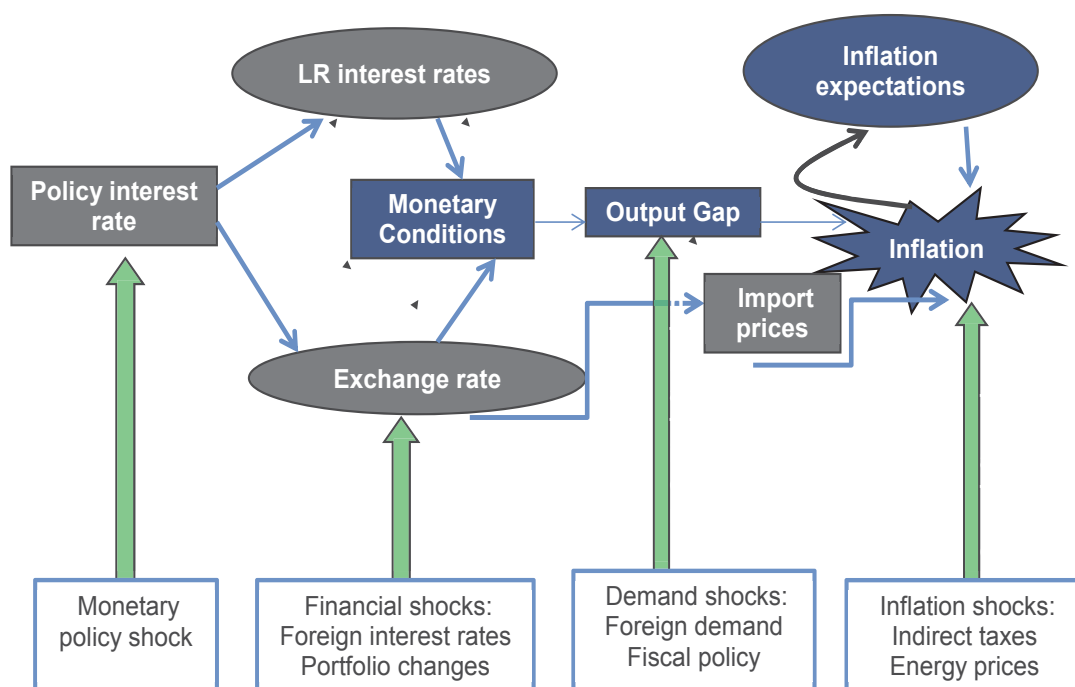
$$i_t^{US} = a_3^{i^{US}} * i_{t-1}^{US} + (1 - a_3^{i^{US}}) * (\bar{r}_{ss}^{US} + \pi_{ss}^{US}) + \varepsilon_t^{i^{US}} \dots\dots\dots(43)$$

$$\pi_t^{US} = a_4^{\pi^{US}} * \pi_{t-1}^{US} + (1 - a_4^{\pi^{US}}) * \pi_{ss}^{US} + \varepsilon_t^{\pi^{US}} \dots\dots\dots(44)$$

2.2 The transmission mechanisms implied by the model

The FPAS macro model for Rwanda assumes policy interest rate as the monetary policy instrument. The channels through which monetary policy affects the real economy are summarized in figure 1 and they include: the interest channel, the asset prices channel, the expectations channel and the exchange rate channel.

Figure 1: Transmission mechanisms



However, for the Rwandan economy, the main interest rate channel, exchange rate channel and expectations channel are more relevant though not fully effective. The interest channel begins with the pass-through from the policy rate to the market rates thereby by affecting the lending behavior of banks which has an impact on aggregate demand (through real monetary conditions index) and inflation. The exchange rate effects feed into inflation either directly or indirectly. The indirect channel works in a way that the change in the policy rate affects capital flows which in turn affects the



exchange rate. The change in exchange rates then affects the cost of imported intermediate goods and then domestic production and inflation. The direct effect occurs when exchange rate movements affect the prices of imported finished goods, directly feeding into inflation. The transmission mechanism in figure (1) takes care of the different shocks that may hit the economy, for example supply shocks in form of increasing/decreasing international commodity prices and domestic food supply shocks.

2.3. Data and calibration

The data on the interbank rate, exchange rate CPI, and GDP used were sourced from the National Bank of Rwanda and the National Institute of statistics of Rwanda, respectively. The CPI data includes the urban consumer price index as well as the core and non-core components. The non-core index represents about 75% of the overall CPI. The non-core index therefore represents 25% of the overall urban index, comprising of 7.8% for the energy index and 17.3% for the fresh products index. The other data used are the bilateral nominal exchange rates (RWF/USD), the interbank rate and real GDP growth rate. The international commodity prices (food & oil) as well as other foreign sector variables were obtained from OGRESEARCH. All the domestic variables are seasonally adjusted, log transformed and converted into quarterly frequency prior to being used. Some of the checks used to guide calibrations include examination of the impulse response functions, in sample simulations and historical decompositions to ensure they make economic sense and correctly portray the Rwandan economic realities, the starting point of course being the experience from other countries. The calibrations of the parameters used in the model are given in the appendix.

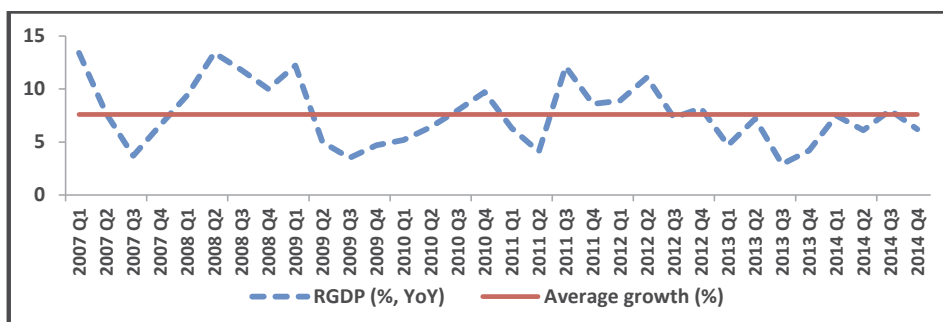
3. A model-based historical interpretation of the Rwandan economy

This section applies the model to Rwandan data with the objective of using the model results (Kalman Filtered results) to explain the key developments in the Rwandan economy over the recent past with respect to the drivers of inflation and real GDP growth dynamics as well as exchange rate movements.

3.1 Growth dynamics

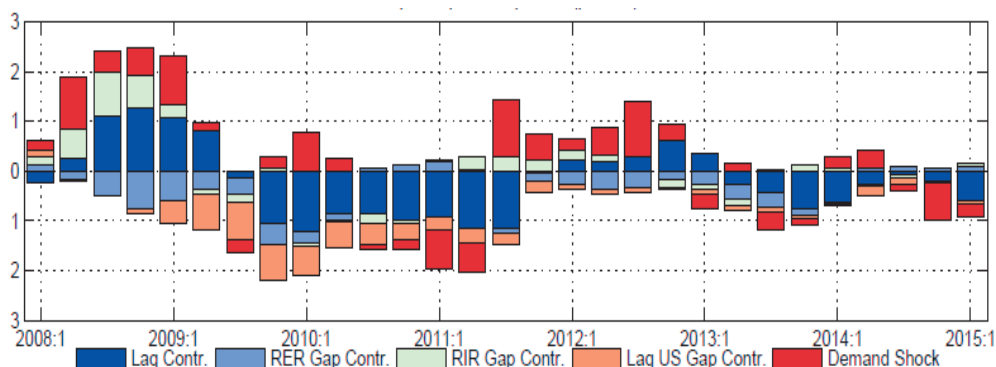
Since 2007, the Rwandan economy has been steadily growing at an average rate of about 7.6% on annual basis on the back of sound macroeconomic policies and higher public investments.

Figure 2: real GDP growth (% , Y-o-Y)



Source: BNR Monetary Policy Department

The model successfully explain the drivers behind the economic performance for key periods such as: the strong growth in 2008, the subsequent but delayed slowdown in 2009 due to the global financial crisis, the economic recovery of 2012 followed by the deceleration in economic performance especially in 2013 on the back of the aid shock and then the economic recovery after the shock. The 2008 increasing real economic activities mainly resulted from the demand shock and easy monetary policy as denoted by the positive contribution of these elements to the positive output gap. This period of double digit growth was in 2008Q2-2009Q1. The demand shock here may refer to expansionary fiscal policy that is not explicitly modeled.

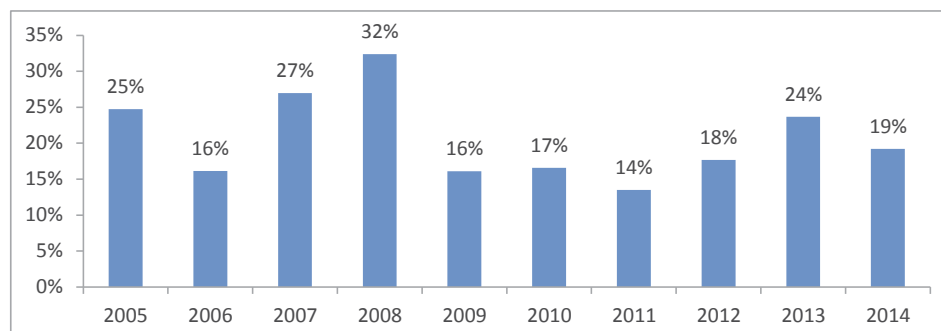
Figure 3: Decomposition of the output gap

Source: BNR Monetary Policy Department

High demand in 2008 was supported by higher fiscal spending as evidenced by the increase in total nominal fiscal expenditure by 32%, the highest since 2005. However, though fiscal spending increased, its nominal value was amplified by the high level of inflation (i.e. 15.4% on annual average) that prevailed at the time.

The period of double digit growth was followed by economic contraction (single-digit growth) in 2009Q2-2011Q2 that was a result of the global financial crisis. The latter led to the decline in the global economy with negative implications on domestic demand, as captured by the negative contribution of the foreign output gap. Towards the end of this episode, the demand shock also dragged down the economy. The economy however recovered in 2012, thanks to the expansionary fiscal policy, leading to the positive contribution of the demand shock to the domestic output gap. In addition, monetary policy was also supportive to growth given the increasing credit to private sector that came as a result of a looser monetary policy that prevailed since 2011. This recovery was however short lived following the negative effects of the aid shock in the second half of 2012 that mainly materialized in 2013 as evidenced by the contraction in the demand shock contribution to the domestic output gap due to the shortfall in both household and government spending.

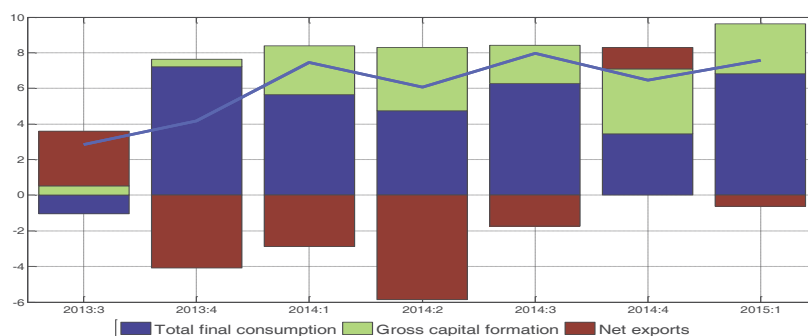
Figure 4: Nominal total fiscal expenditure and net lending (% , y-o-y)



Source: MINECOFIN

In 2014Q4 (figure 5), the economy started to slowdown following a decline in total final consumption in particular and aggregate demand in general. In the first quarter of 2015, the economy recovered, with real GDP growing by 7.6% following a rebound in total final consumption.

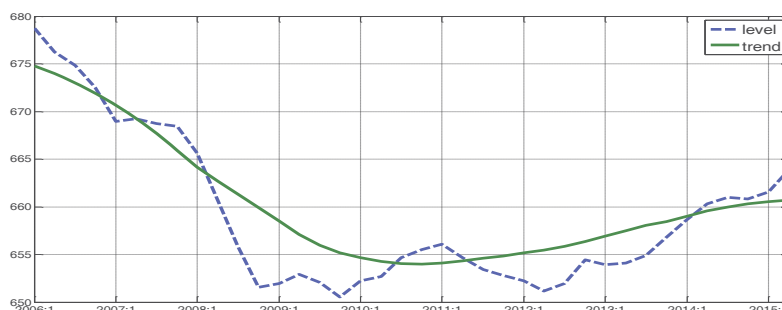
Figure 5: Real GDP growth in Rwanda



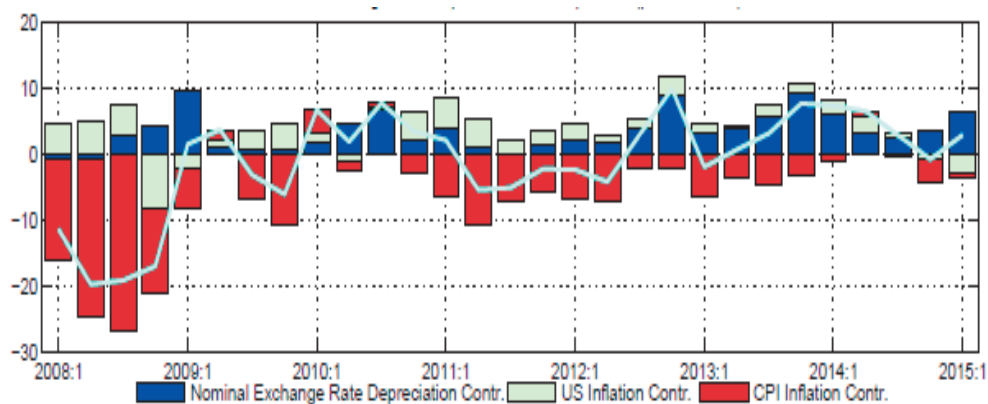
Source: BNR Monetary Policy Department

3.2 Exchange rate developments

The exchange rate dynamics since 2008 can be split into three episodes: the first episode is in 2008-2010 where the real exchange (RER) was appreciating, followed by the relative RER stability episode in mid-2010-mid 2011 and then the gradual RER depreciation thereafter, as shown in figure (6).

Figure 6: Real Exchange rate depreciation: FRW/USD**Source:** BNR Monetary Policy Department

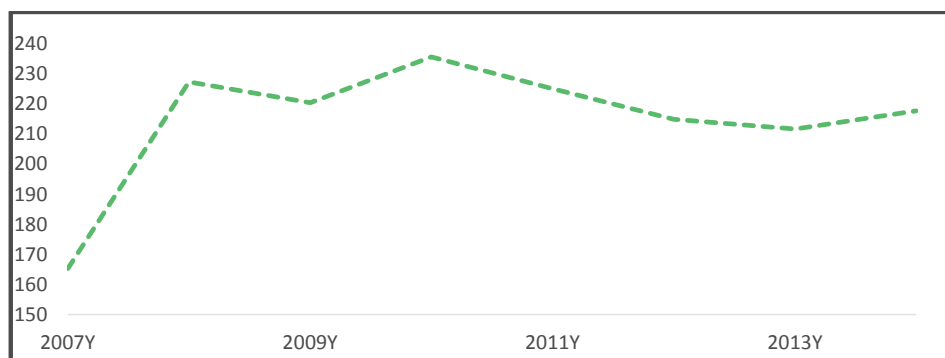
During the first episode, the RER appreciation was mainly driven by the fast increasing domestic inflation relative to foreign inflation, consistent with the fact that in 2008 inflation was rising on the account of rising international commodity prices, especially oil. The exception was in 2008Q4-2009Q1 where foreign inflation also contributed to RER appreciation and in 2008Q3-2009Q1 when the nominal exchange rate contributed to the slowdown in the RER appreciation.

Figure 7: Decomposition of Real Exchange Rate depreciation**Source:** BNR Monetary Policy Department

The second episode was characterized by the relative stability in the RER owing to the slowdown in domestic inflation, the relative stability of foreign inflation at around 2%

and stability of the FRW against the USD. The third and most recent episode was generally driven by the nominal depreciation of the FRW against the USD following the shortfall in aid inflows in the second half of 2012 and most recently, by the strengthening of the USD following the good performance of the US economy. This period coincided with moderate domestic and global inflation. Given the persistent nature of the US dollar and the expected monetary policy tightening towards the end of 2015, the ongoing depreciation of the FRW against the USD is also likely to continue in the medium term (Druck et al., 2015). The exchange dynamics during the three episodes are tracked quite well by the trend of the terms of trade index as shown in figure 8.

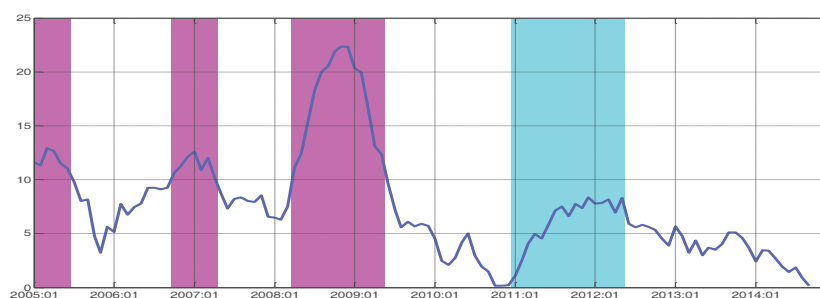
Figure 8: Terms of Trade (2000Y=100)



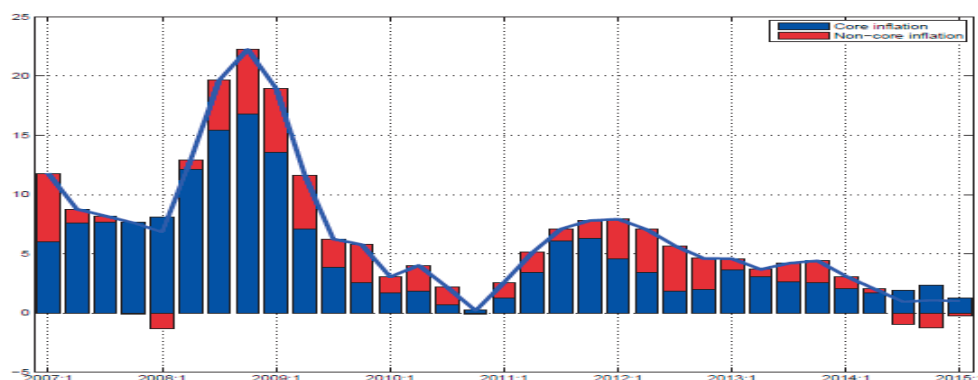
Source: World Bank & Bloomberg

3.3 Inflation developments

The key inflation episodes in Rwanda typically fall in three episodes, consistent with real economic growth and exchange rate developments discussed before. The first episode covers the 2008H2-2009H1 when inflation was in double digit, followed by the second episode of 2011 when inflation was rising but still in single digit and the third episode of 2014-15 characterized by falling inflation. These periods are chosen because of well-known developments in international commodity prices.

Figure 9: Key episodes for headline inflation**Source:** BNR Monetary Policy Department

The role of international commodity prices can be viewed by looking at the contribution of non-core inflation to overall inflation. In the first and second episode, the contribution of non-core inflation was quite high, explaining why monetary authorities were reluctant to tighten the policy stance as the model suggests. In the last episode, the international commodity prices are declining, explaining why the contribution of non-core inflation is negative.

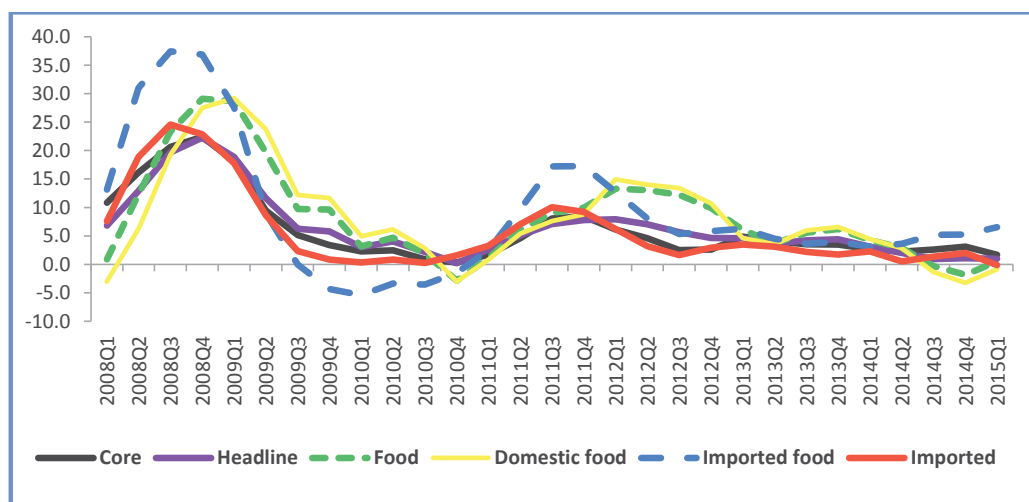
Figure 10: Decomposition of headline inflation**Source:** BNR Monetary Policy Department

In the explanation of inflation dynamics, the model was designed to separate core from non-core inflation dynamics so as to be able to dissect the role of international

commodity prices, especially oil. The model detects the effect of imported food inflation on core inflation via the direct channel while the effect of imported oil inflation is captured through the indirect channel. Conversely the effect of both imported oil and food inflation affect non-core inflation through the indirect channel.

During the early period of the first episode, core inflation was rising mainly on the account of accelerating domestic food inflation (the positive contribution of the supply shocks) and marginally due to the influence of rising imported food inflation as well as increasing oil inflation (positive RMC contribution). Despite good agricultural harvests (total food production rose by 19.2% in 2008 and 12.6% in 2009), food inflation remained high following higher imported food inflation.

Figure 11: The influence of domestic and imported food inflation



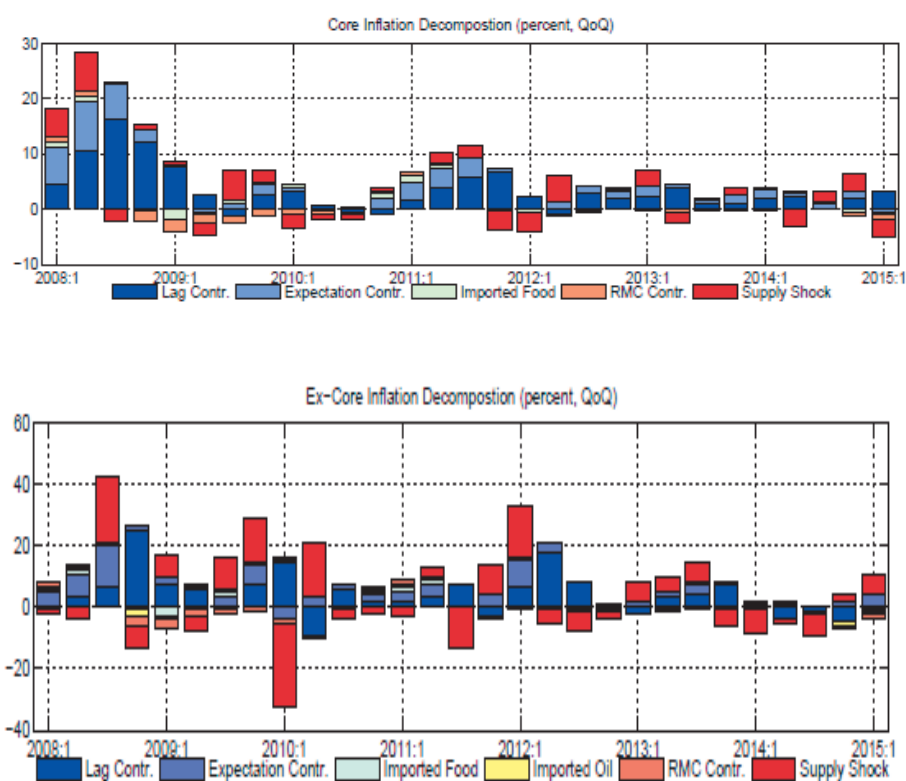
Source: BNR Monetary Policy Department

Owing to the rising trend of inflation globally, the role of expectations is also quite pronounced. Note that the contribution of monetary policy is missing due to the fact the model is designed in such a way that monetary policy reacts to headline inflation.

In the 2011 inflationary episode, the role of imported food inflation is slightly important at the beginning but fades later while the impact of supply shocks and

inflation expectations grow over time. This is consistent with the fact that the EAC region was experiencing droughts and rising food inflation in tandem with rising international commodity prices. The last episode starts in 2014 and is characterized by the decelerating international commodity prices, though their effect is rather small but likely to increase following the recent upward revision of local pump prices from 810 FRW/liter to 840 FRW/liter. Since core CPI account for about 78% of the overall CPI, the developments in core inflation tend to shape headline inflation in general.

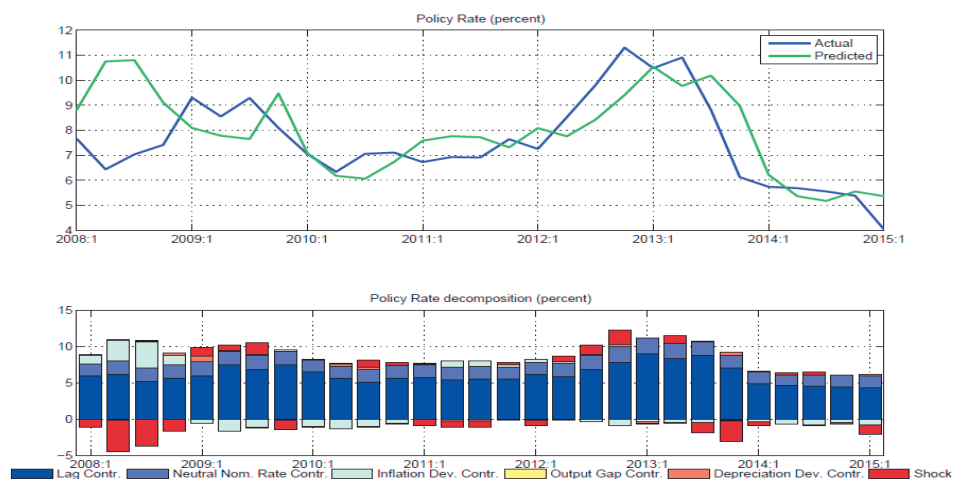
Figure 12: Decomposition of core and non-core inflation



Source: BNR Monetary Policy Department

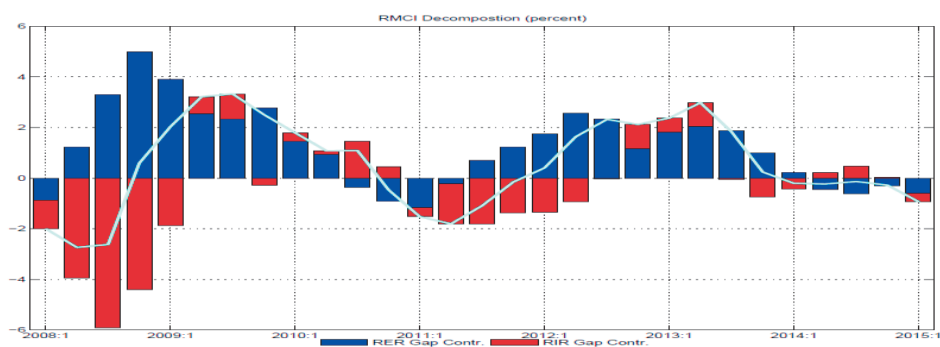
As mentioned earlier, the interbank rate is used here as the policy rate, especially because it reflects liquidity conditions and is also closely linked to other interest rates.

Figure 14: Monetary policy decomposition



Towards, the end of the last episode (2014-15), the National Bank of Rwanda adopted a loose monetary policy on the back of persistently declining inflation and the need to support economic recovery. This corresponds to the negative contribution of the real interest gap to the real monetary condition index as shown in figure 15.

Figure 15: Decomposition of the Real Monetary Conditions index (RMCI)



Source: BNR Monetary Policy Department



4. Conclusion

The FPAS macro-model discussed in this paper is designed to capture the key features of the Rwandan economy and will be a key building block for the transition to a forward-looking price based monetary policy framework. Despite its simplicity, the model tracks well the key developments in the Rwandan economy notably the inflationary periods of 2008-09 and 2011, the deflationary period of 2014-15 and the movements in the real exchange rates: appreciation in 2007-2010, stability thereafter followed by mild RER depreciation driven by the strengthening of the USD against the FRW. The model however has some drawbacks that need to be addressed in future by for example including the fiscal sector, incorporating the policy actions, especially exchange rate smoothing, and the role of money. Basing on the good performance of the model, the National Bank of Rwanda has started using some of its results, such as inflation and output gap projections, to inform monetary policy decision making. After making the mentioned improvements, the model will be fully operationalized to produce forecasts on key macro-economic variables and to provide a consistent framework for monetary policy discussions.

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APPENDIX

Appendix 1: Calibration of the model parameters

Parameter	Description	Value
Output gap equation		
a_1^y	AR(1) parameter; output gap persistence	0.65
a_2^y	Coefficient of the real monetary conditions	0.15
a_3^y	Coefficient of the foreign output gap	0.15
a_1^{rmi}	Weight of the real exchange rate in the real monetary conditions	0.6
Core inflation equation		
b_1^{core}	AR(1) parameter; core inflation persistence	0.58
b_2^{core}	Coefficient of the real marginal costs in core inflation	0.4
b_3^{core}	Coefficient of international food price pressures adjusted by relative prices	0.02
$b_1^{rmi_core}$	Weight of output gap in real marginal costs	0.50
$b_2^{rmi_core}$	Weight of imported food prices gap and exchange rate gap in real marginal costs adjusted by relative price gap	0.05
$b_3^{rmi_core}$	Weight of imported oil prices gap and exchange rate gap in real marginal costs adjusted by relative price gap	0.07
Non-core inflation equation		
b_1^{xcore}	AR(1) parameter; non-core inflation persistence	0.55
b_2^{xcore}	Coefficient of the real marginal costs non-core inflation	0.60
b_3^{xcore}	Coefficient of international food prices pressures adjusted by relative prices	0.03
b_4^{xcore}	Coefficient of international oil prices pressures adjusted by relative prices	0.01
$b_1^{rmi_xcore}$	Weight of output gap in real marginal costs	0.55
$b_2^{rmi_xcore}$	Weight of imported food prices gap and exchange rate gap in real marginal costs adjusted by relative price gap	0.03

$b_3^{\text{rnc_xcore}}$	Weight of imported oil prices gap and exchange rate gap in real marginal costs adjusted by relative price gap	0.07
Headline inflation equation		
wl^{core}	Weight of core inflation in Headline	0.7485
Exchange rate equation (UIP)		
d_1	AR(1) parameter; coefficient of the expected exchange rate	0.6
Monetary policy rule		
f_1	AR(1) parameter; Policy rate persistence	0.8
f_2	Coefficient of the expected inflation deviations from the target	1.5
f_3	Coefficient of the output gap in monetary policy rule	0.2
f_4	Coefficient of the exchange rate deviations from the long run trend	0.4
Oil inflation equation		
$c_1^{\text{Al_qoil_tnd}}$	AR(1) parameter; coefficient of the change in oil prices trend	0.9
$c_1^{\text{l_qoil_gap}}$	AR(1) parameter; coefficient of the oil prices gap persistence	0.75
Food inflation equation		
$c_1^{\text{Al_qfood_tnd}}$	AR(1) parameter; coefficient of the change in food prices trend	0.9
$c_1^{\text{l_qfood_gap}}$	AR(1) parameter; coefficient of the food prices gap persistence	0.75
Trends		
ϕ_1	Persistence, long run real interest rate	0.8
ϕ_2	Persistence, long run output growth	0.7
ϕ_3	Persistence, long run real exchange rate	0.8
Foreign block		
a_1^{us}	AR(1) parameter, persistence of US output gap	0.80
a_2^{us}	AR(1) parameter, persistence of US real interest rate trend	0.90
a_3^{us}	AR(1) parameter, persistence of US nominal interest rate	0.94
a_4^{us}	AR(1) parameter, persistence of US CPI inflation	0.30

Steady states: long run values		
$\bar{r}_{ss}$	Long run value of the domestic real interest rate trend	4
$\Delta \bar{y}_{ss}$	Long run value of the real GDP growth trend	$100 \cdot \log(1 + 7.5/100)$
$\Delta \bar{z}_{ss}$	Long run value of the real exchange rate trend	-1
$\pi_t^{yoy, tar}$	Domestic CPI inflation steady state/target	$100 \cdot \log(1 + 5/100)$
$\bar{r}_{ss}^{us}$	Long run value of the US real interest rate trend	1
π_{ss}^{us}	Foreign inflation steady state/target	$100 \cdot \log(1 + 2/100)$
Standard deviation of the shocks		
ε_t^y	Shock to domestic output gap	0.9
ε^{core}	Shock to CPI core inflation	1.7
ε^{xcore}	Shock to CPI non-core inflation	5.5
ε_t^s	Shock to nominal exchange rate	1.3
$\varepsilon_t^{\Delta \bar{y}}$	Shock to real GDP growth trend	0.8
$\varepsilon_t^{\Delta \bar{z}}$	Shock to real exchange rate trend	1.5
$\varepsilon_t^{\bar{r}}$	Shock to real interest rate trend	0.7
ε_t^i	Shock to monetary policy rate	0.9
$\varepsilon_t^{\Delta l_{qoil_tnd}}$	Real international oil price trend shock	1
$\varepsilon_t^{\Delta l_{qoil_gap}}$	Real international oil price gap shock	12
$\varepsilon_t^{\Delta l_{qfood_tnd}}$	Real international food price trend shock	1
$\varepsilon_t^{\Delta l_{qfood_gap}}$	Real international food price gap shock	4
$\varepsilon_t^{y^{us}}$	Shock to foreign (US) output gap	0.56
$\varepsilon_t^{\bar{r}^{us}}$	Shock to foreign(US) real interest rate trend	0.1
$\varepsilon_t^{i^{us}}$	Shock to foreign(US) nominal interest rate	0.44
$\varepsilon_t^{\pi_t^{us}}$	Shock to foreign(US) CPI inflation	2.03

Source: Monetary Policy & Research Department